

CSE 582 Bioinformatics Final

QUESTIONS

1. The following algorithm is a cousin of one of the motif finding algorithms. Identify which algorithm is a cousin of ANOTHERMOTIFSEARCH and find the similarities and differences between these two algorithms.

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ANOTHERMOTIFSEARCH(DNA, t, n, l)
1 s = (1, 1, . . . , 1)
2 bestMotif = FINDINSEQ(s, 1, t, n, l)
3 return bestMotif
FINDINSEQ(s, currentSeq, t, n, l)
1 bestScore = 0
2 for j = 1 to n - l + 1
3     s(currentSeq) = j
4     if currentSeq != t
5         s = FINDINSEQ(s, currentSeq+1,
                        t, n, l)
6     if Score(s) > bestScore
7         bestScore = Score(s)
8         bestMotif = s
9 return bestMotif

```

2. Design an input for the GREEDYMOTIFSEARCH algorithm that causes the algorithm to output an incorrect result. That is, create a sample that has a strong pattern that is missed because of the greedy nature of the algorithm. If optimalScore is the score of the strongest motif in the sample and greedyScore is the score returned by GREEDYMOTIFSEARCH, how large can optimalScore/greedyScore be?

3. What is the optimal global alignment for APPLE and HAPPE? Show all optimal alignments and the corresponding paths under the match premium +1, mismatch penalty -1, and indel penalty -1.

4. The SBH problem is to reconstruct a DNA sequence from its l -mer composition

- Suppose that instead of a single target DNA fragment, we have two target DNA fragments and we simultaneously analyze both of them with a universal DNA array. Give a precise formulation of the resulting problem (something like the formulation of the SBH problem).
- Give an approach to the above problem which resembles the Hamiltonian Path approach to SBH.
- Give an approach to the above problem which resembles the Eulerian Path approach to SBH.

5. Determine the number of different ways to partition a set of n elements into k clusters.

6. Suppose a dishonest dealer has two coins, one fair and one biased; the biased coin has heads probability $1/4$. Assume that the dealer never switches the coins. Which coin is more likely to have generated the sequence HTTTHHHTTTTTHHTT? ($\log_2(3) = 1.585$).